

Rearranging the terms and removing repeated terms.
 $(A+B+C+D) \cdot (A+B+C+D) \cdot (A+B+C+D) \cdot (A+B+C+D) \cdot (A+B+C+D)$
 $(A+B+C+D) \cdot (A+B+C+D) \cdot (A+B+C+D) \cdot (A+B+C+D)$

$\Pi(1, 3, 4, 9, 11, 12, 13, 14, 15)$

Simplified POS:

$$F = (B+D) \cdot (A+B) \cdot (B+C+D)$$

AB \ CD	00	01	11	10
00		0	0	
01	0			
11	0	0	0	0
10		0	0	

$$F' = (B+D) \cdot (A+B+D) \cdot (A+B+C)$$

$$F = (F')' = ((B+D) \cdot (A+B+D) \cdot (A+B+C))'$$

AB \ CD	00	01	11	10
00	0			0
01		0	0	0
11				
10	0			0

$$= (B+D)' + (A+B+D)' + (A+B+C)'$$

$$= B'D' + A'B'D' + A'B'C' \quad (\text{Extended Theorem 5(b)})$$

$$= B'D' + A'BD + A'BC \quad (\text{Theorem 3})$$

3-9: Simplify each of the following functions and implement them with NAND gates.

(b) $F_2 = (A+C) \cdot (A+B+C) \cdot (A+B+C+D) \cdot (B+C+D)$ (Postulate 2(a))
 $= (A+C) \cdot (A+B+C) \cdot (A+B+C+D) \cdot (B+C+D)$ (Postulate 5(b))
 $= (A+C+B \cdot B') \cdot (A+B+C) \cdot (A+B+C+D) \cdot (B+C+D)$ (Postulate 4(b))
 $= (A+C+B) \cdot (A+C+B') \cdot (A+B+C) \cdot (A+B+C+D) \cdot (B+C+D)$ (Postulate 2(a))
 $= (A+C+B+0) \cdot (A+C+B+0) \cdot (A+B+C+0) \cdot (A+B+C+D) \cdot (B+C+D+0)$ (Postulate 2(a))
 $= (A+C+B+D \cdot D') \cdot (A+C+B+D \cdot D') \cdot (A+B+C+D \cdot D')$ (Postulate 5(b))
 $= (A+C+B+D) \cdot (B+C+D+A \cdot A') \cdot (A+C+B+D) \cdot (A+C+B+D) \cdot (A+B+C+D)$
 $= (A+C+B+D) \cdot (A+C+B+D) \cdot (A+C+B+D) \cdot (A+B+C+D) \cdot (B+C+D+A)$